Restaurant Menu and Ordering System using Angular and TypeScript

1. Introduction

The **Restaurant Menu and Ordering System** is a modular web application developed using the **Angular** framework and **TypeScript**. This application is designed to provide customers with a seamless digital experience for browsing dishes, viewing detailed menu information, managing a virtual shopping cart, and placing orders. The project serves as a practical implementation of advanced Angular concepts, including reactive forms, state management through services, and asynchronous data handling.

2. Objective

The primary objectives of this project are to:

- Design an interactive digital menu for a restaurant environment.
- Implement a robust cart management system for selecting and managing items.
- Utilize **Angular Material** to build a professional, responsive user interface.
- Apply advanced architectural concepts such as **Lazy Loading** and **Route Guards**.
- Master data-driven development using **RxJS Observables** and **HttpClient**.

3. Tools and Technologies Used

- **Framework:** Angular (v17+)
- **Language:** TypeScript
- **UI Library:** Angular Material (MatCard, MatTable, MatToolbar, MatDialog)
- **Backend:** Mock JSON Server / Static JSON files for menu and order data
- **Tools:** Node.js, Angular CLI, Visual Studio Code, RxJS

4. System Architecture

The application follows a strictly modular, component-based architecture to ensure scalability and ease of maintenance.

Main Components:

- **Navbar Component:** Provides global navigation across the menu, cart, and checkout.
- **Menu List Component:** Displays all available dishes categorized by type (e.g., Appetizers, Main Course).
- **Menu Detail Component:** Uses dynamic routing to show specific item descriptions, ingredients, and pricing.
- **Cart Component:** Manages the state of selected items, allowing users to update quantities or remove dishes.
- **Order Form Component:** A complex reactive form for capturing customer details and payment info.

5. Implementation Details

5.1 Setup and Fundamentals

The environment was initialized using **Angular CLI**. Data models were strictly typed using TypeScript **interfaces** (e.g., MenuItem, Order) to ensure type safety across the application.

5.2 Services and Data Management

- **MenuService:** Handles HTTP requests to fetch menu data from the JSON backend.
- **CartService:** Uses a centralized state to track items added by the user, shared across the app via **Dependency Injection**.

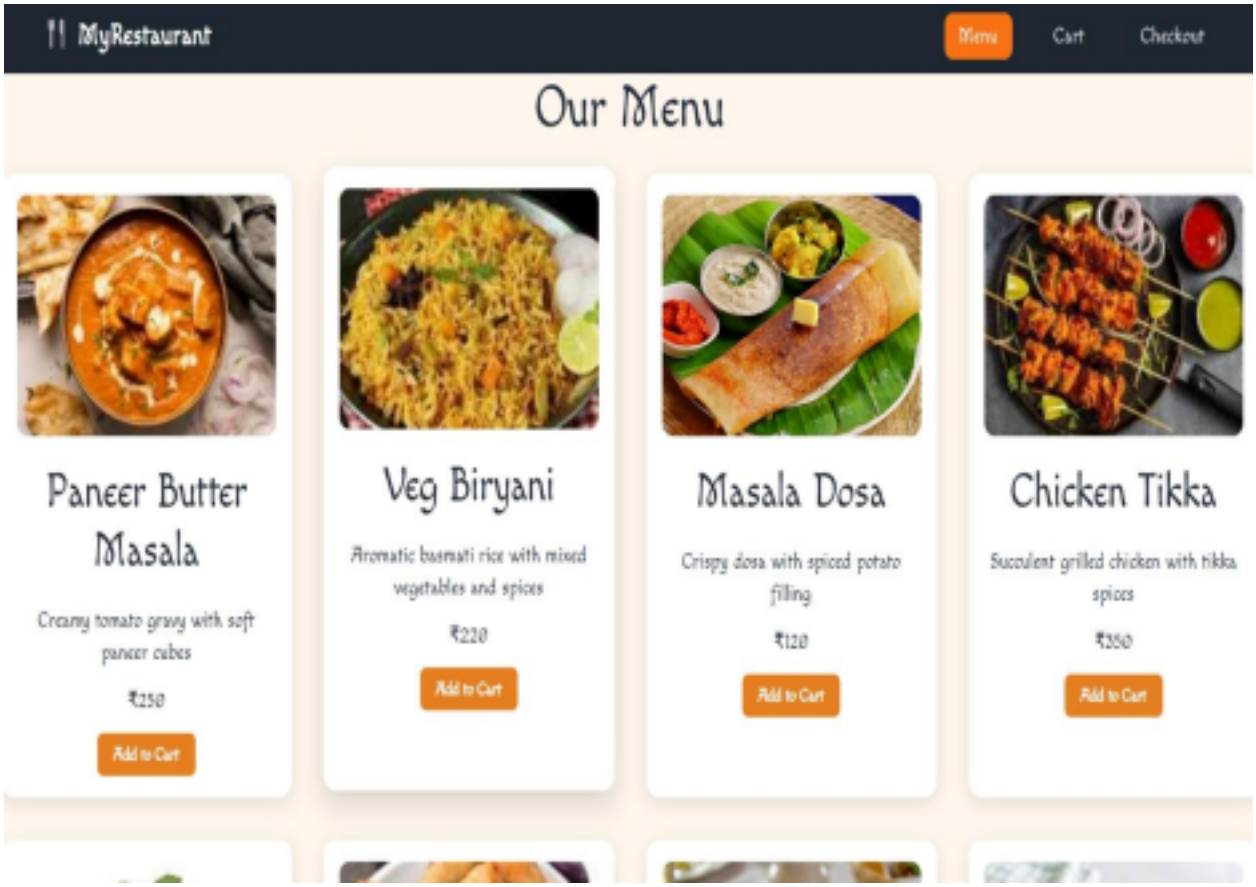
5.3 Angular Features Used

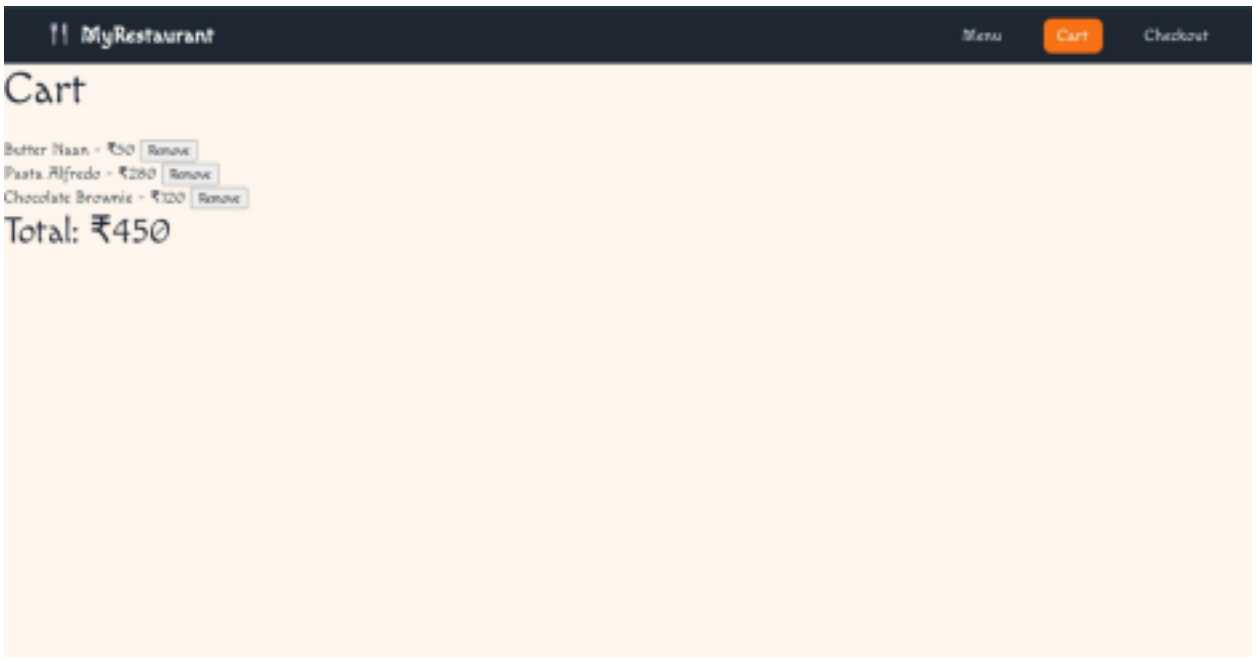
- **Reactive Forms:** Employed for the checkout process with built-in and custom validations.
- **Routing & Guards:** Configured routes for /menu, /cart, and /checkout. **Route Guards** prevent users from accessing the checkout page if the cart is empty.
- **Custom Pipes:** Created to filter menu items by price range or category dynamically.
- **Directives:** Used structural directives (*ngIf, *ngFor) for dynamic rendering and attribute directives for highlighting discounted items.

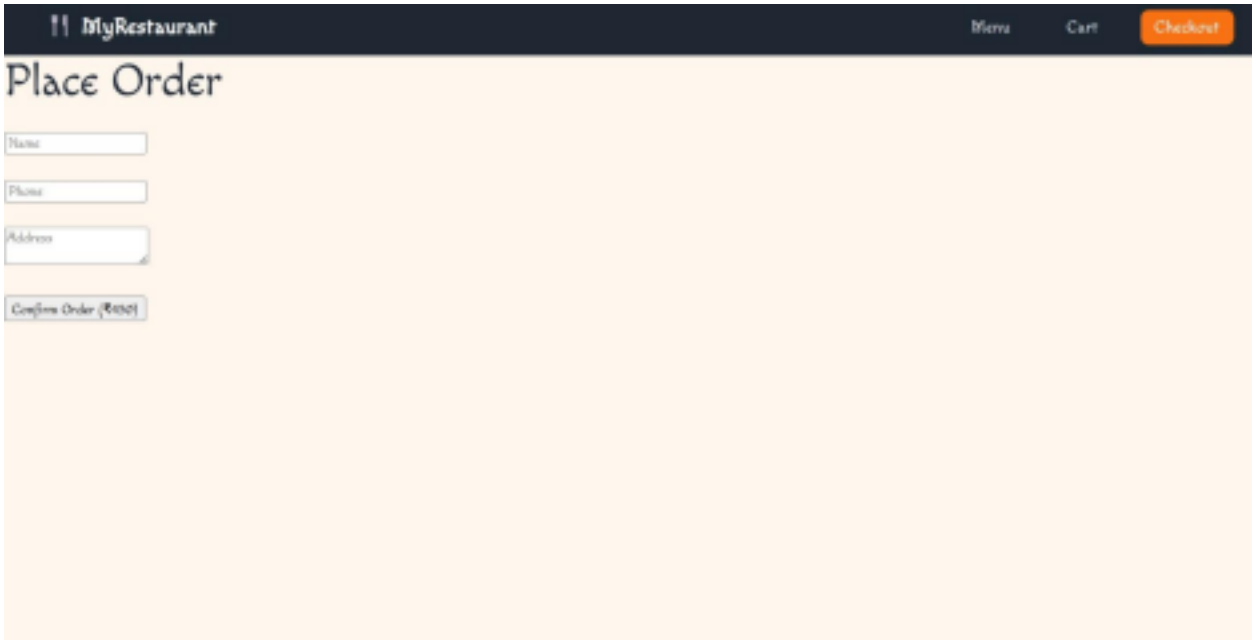
6. Conclusion

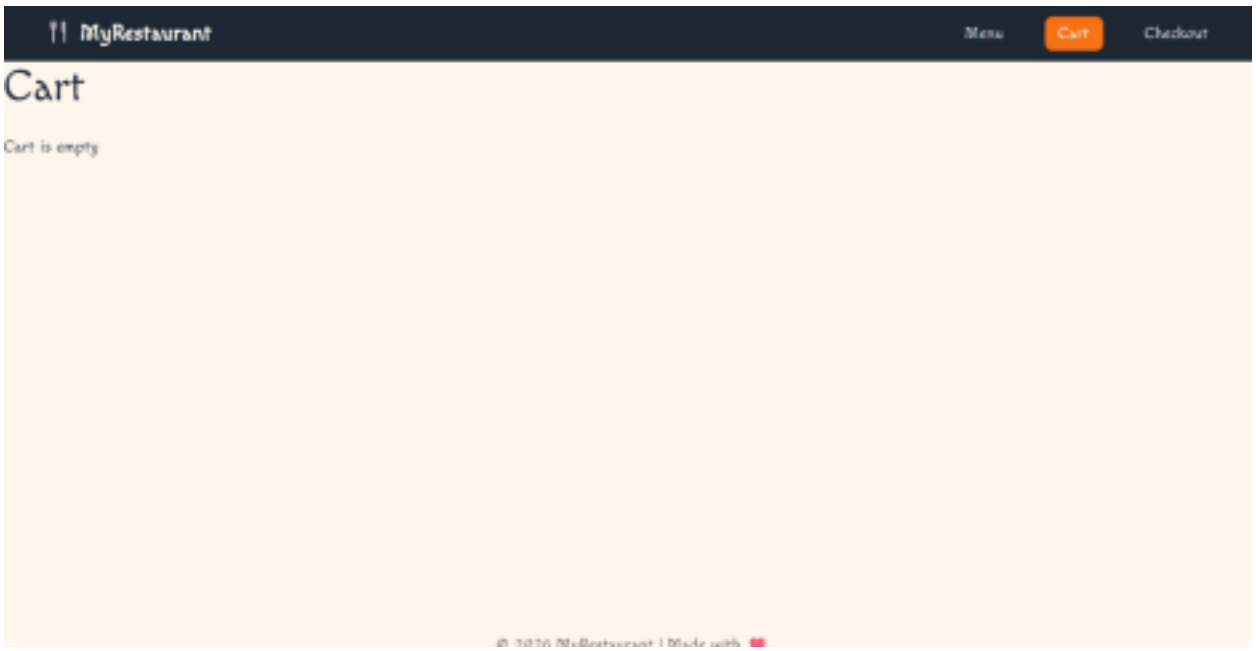
The Restaurant Menu and Ordering System demonstrates the power of Angular in building data-driven, real-world applications. By integrating **Angular Material** for design and **RxJS** for asynchronous operations, the project provides a high-performance, responsive solution that mirrors modern e-commerce and hospitality platforms.

7. Reference Screenshort:




MyRestaurant
Menu
Cart
Checkout
Cart
Butter Naan - ₹50 Remove
Pasta Alfredo - ₹280 Remove
Chocolate Brownie - ₹120 Remove
Total: ₹450


MyRestaurant
Menu
Cart
Checkout
Place Order
Name
Phone
Address
Confirm Order (₹450)



JERUSHA S-2462082

SHERYN ANAND-2462148

ALAN GEORGE JOSEPH-2462023

NEVAN MIRANDA-2462120